e) $\frac{Q^2}{g} = \frac{A^3}{B}$

$A_c = h_c(b + mh_c) = h_c(3.88 + 2h_c)$

$B = b + 2mh_c = 3.88 + 4h_c$

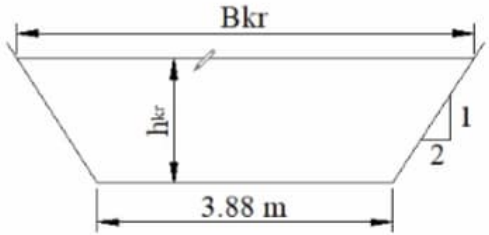


Figure p8.2f

$\frac{12^2}{9.81} = \frac{[h_c(3.88 + 2h_c)]^3}{(3.88 + 4h_c)}$ or $14.68 = \frac{[h_c(3.88 + 2h_c)]^3}{(3.88 + 4h_c)}$

by trial and error $h_c = 0.85m$

$h = 0.97 > h_c = 0.85$ sub-critical regime

Critical velocity;

$$V_c = \frac{Q}{A_c}$$

$$A_c = (3.88 + 2 \times 0.85)0.85 = 4.74 m^2$$

$$V_c \frac{12}{4.74} = 2.53 m/s$$

Uniform flow velocity;

$$V = \frac{Q}{A} = \frac{12}{6 \times 0.97^2} = 2.12 \text{ m/s}$$

$V < V_c$ NR

Critical slope;

$$V_c = \frac{1}{n} R_c^{2/3} J_c^{1/2}$$

$$R_c = \frac{A_c}{P_c} = \frac{4.74}{3.88 + 2 \times 0.85\sqrt{1+4}} = 0.62 \text{ m}$$

$$2.53 = \frac{1}{0.02} (0.62)^{0.667} J_c^{0.5}$$

$J_c = 0.0048$

$J < J_c$ NR

f) For trapezoidal channel

$$h_m = \frac{A}{B} = \frac{6 \times (0.97)^2}{3.88 + 4 \times 0.97} = 0.74\,\text{m}$$

$$V = \frac{Q}{A} = \frac{12}{5.65} = 2.12\,\text{m/s}$$

$$Fr = \frac{V}{\sqrt{g h_m}} = \frac{2.12}{\sqrt{9.81 \times 0.74}} = 0.74 < 1 \text{ Sub-critical flow}$$

for rectangular channel

$$h_m = h = 1.145\,\text{m}$$

$$A = bh = 1.145 \times 4.58 = 5.24\,\text{m}^2$$

$$V = \frac{12}{5.24} = 2.29\,m/s$$

$$Fr = \frac{V}{\sqrt{gh}} = \frac{2.29}{\sqrt{9.81 \times 1.145}} = 0.68 < 1 \text{ Sub-critical flow}$$

using general equation

$$\frac{Q^2}{g} = \frac{A^3}{B} \text{ or } \frac{Q^2}{g} = \frac{b^3 h_c^3}{b} \text{ and with definition unit discharge } b = \frac{Q}{b}$$

$$h_{kr} = \sqrt[3]{\frac{q^2}{g}} = \sqrt[3]{\frac{\left(\frac{12}{4.58}\right)^2}{9.81}} = 0.89 < h = 1.145\text{m sub critical flow}$$

For triangular channel

$$h_m = \frac{h}{2} = \frac{1.64}{2} = 0.82\,\text{m}$$

$$A = 6.56 \times 1.64 / 2 = 5.38\,\text{m}^2$$

$$V = \frac{12}{5.38} = 2.23\,\text{m/s}$$

$$Fr = \frac{2.23}{\sqrt{9.81 \times 0.82}} = 0.79 < 1$$

sub-critical flow

g) For hydraulically economic

$$b = 2h(\sqrt{1+m^2} - m)$$

$$b = 2h(\sqrt{1+4} - 2)$$

$$b = 0.472h$$

$$A = h(b + mh) = h(0.472h + 2h) = 2.472h^2$$

$$h = 1.475\,m \qquad b = 0.696\,m$$

h) $$\frac{Q^2}{g} = \frac{A^3}{B}$$

$$A_c = h_c(b + mh_c)$$

$$A_c = h_c(0.696 + 2h_c)$$

Problem 8.3: At he cross-section of the flow in a river during a flood was shown in Fig.p8.3. Assuming the roughness coefficient for the side channel and main channel n= 0.025. Bed slope is 0.0007.Calculate the discharge in the channel.

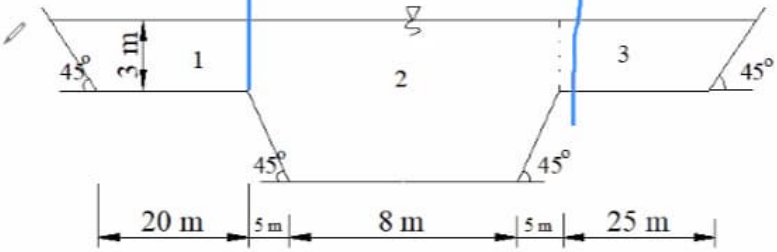


Figure 8.3

Solution:

$$A_1 = \frac{3x3}{2} + 3x20 = 64.5\ m^2$$

$$A_2 = (5+8+5)\times 3 + [8+(5+8+5)]\times \frac{5}{2} = 119\ m^2$$

$$A_3 = \frac{3x3}{2} + 3x25 = 79.5 \text{ m}^2$$

$$P_1 = 20 + \sqrt{3^2 + 3^2} = 24.24 \text{ m}$$

$$P_2 = 8 + 2 \times \sqrt{5^2 + 5^2} = 22.14 \text{ m}$$

$$P_3 = 25 + \sqrt{3^2 + 3^2} = 29.24 \text{ m}$$

$$R_1 = \frac{A_1}{P_1} = \frac{64.5}{24.24} = 2.66 \text{ m}$$

$$R_2 = \frac{A_2}{P_2} = \frac{119}{22.14} = 5.37 \text{ m}$$

$$R_3 = \frac{A_3}{P_3} = \frac{79.5}{29.24} = 2.72 \text{ m}$$

$$Q_1 = \frac{1}{n} R_1^{2/3} J_0^{1/2} A_1 = \frac{1}{0.025} \times (2.66)^{2/3} \times 0.0007^{1/2} \times 64.5 = 131.05 \text{ m}^3/\text{s}$$

$$Q_2 = \frac{1}{n} R_2^{2/3} J_0^{1/2} A_2 = \frac{1}{0.025} \times (5.37)^{2/3} \times 0.0007^{1/2} \times 119 = 386.19 \text{ m}^3/\text{s}$$

$$Q_3 = \frac{1}{n} R_3^{2/3} J_0^{1/2} A_3 = \frac{1}{0.025} \times (2.72)^{2/3} \times 0.0007^{1/2} \times 79.5 = 163.94 \text{ m}^3/\text{s}$$

$$Q = Q_1 + Q_2 + Q_3 = 681.18\ m^3/s$$

Problem 8.4: At the cross-section of the flow in a concrete channel was shown in Fig.8.4. Assuming the roughness coefficient for the channel n= 0.012. Bed slope is 2m/km.Calculate the discharge in the channel.

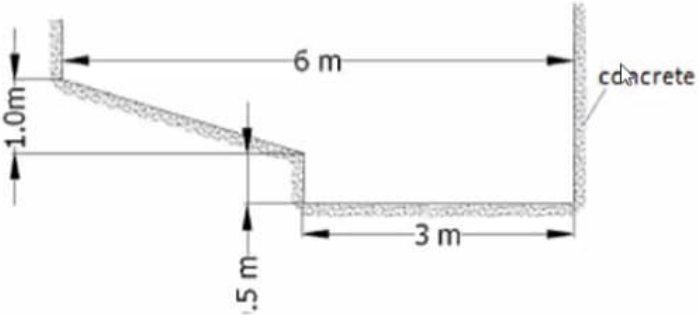


Figure 8.4

R_1=(3x1/2)/3.16=0.47 m R_2=(1.5x3)/5=0.9 m

$$Q_1 = \frac{1}{0.012} \times 0.47^{2/3} \times 0.002^{1/2} \times 1.5 = 3.38\ m^3/s$$

$$Q_2 = \frac{1}{0.012} \times 0.9^{2/3} \times 0.002^{1/2} \times 4.5 = 15.63\ m^3/s$$

Q_t=19.01 m^3/s